

Joint Score of a Diffusion Model Applied to an AR(1) Sequence

A Self-Contained Derivation

Giovanni Mantovani & Marco Lomele — MSc Thesis, Bocconi University,
2025–2026

Supervisors: Prof. Marc Mézard Tutor: Jérôme Garnier

Contents

1	Motivation and Problem Statement	2
2	The AR(1) Gaussian Model	3
2.1	Model specification	3
2.2	Moments of the chain	3
3	The Clean Joint Distribution P_0	4
3.1	Markov factorisation	4
3.2	The precision matrix	5
4	The Noisy Joint Distribution P_t	6
4.1	The Ornstein-Uhlenbeck forward process	6
4.2	The marginal distribution of noisy observations	6
4.3	Method 1: Linear transformation of Gaussians	7
4.4	Method 2: Direct Gaussian integral	7
5	The Joint Score Function	9
5.1	Derivation	9
5.2	The k -th component and inter-frame coupling	10
6	Connection to Mézard’s Formula	11
6.1	Equivalence of the two formulas	12
7	Summary and Notation Table	13
7.1	The complete derivation chain	13
7.2	Key formulas collected	13
7.3	Notation reference	13
8	Discussion: Alternative Noise Models for Sequences	14
8.1	What does it mean to apply diffusion noise to a sequence?	14
8.2	Case 1: i.i.d. OU per frame (our setting)	14
8.3	Case 2: correlated OU noise across frames	15
8.4	Case 3: a single shared noise applied to the full sequence	15
8.5	What if P_0 is not Gaussian?	16

9	Extension to Multivariate Frames	16
9.1	Setup	16
9.2	Does $P_0(\mathbf{a})$ remain Gaussian?	16
9.3	The covariance matrix in the multivariate case	17
9.4	The precision matrix: block-tridiagonal structure	17
9.5	The OU diffusion and joint score in the multivariate case	17
10	The Kalman Filter and Smoother	18
10.1	Why the Kalman smoother matters here	18
10.2	The Kalman filter: forward pass	18
10.3	The Kalman smoother: backward pass	19
10.4	Connection to the joint score	19
11	Suggested Visualisations	19
11.1	Static plots	19
11.2	Animations	20

1 Motivation and Problem Statement

Standard score-based diffusion models generate *static* objects (single images). This work studies the *dynamic* setting: a **dynamic object** is a sequence of $K + 1$ frames

$$a_0, a_1, \dots, a_K \in \mathbb{R}$$

indexed by *internal time* $k = 0, 1, \dots, K$, where consecutive frames are correlated through a known Markov dynamics.

The naive approach embeds the full trajectory in $\mathbb{R}^{K+1}$ and learns a single score function — wasteful, because consecutive frames are highly correlated. Our goal is to understand and exploit the **temporal structure of the joint score field**

$$\mathbf{S}(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log P_t(x_0, x_1, \dots, x_K),$$

where t is the diffusion (OU noise) time applied *independently* per frame, and $\mathbf{x} = (x_0, \dots, x_K)$ are the noisy observations.

We pursue this within the **AR(1) Gaussian toy model**, which is fully solvable in closed form: every distribution is Gaussian, every expectation is computable, and the Kalman smoother provides the exact posterior.

Critical distinction. Previous work computed the per-frame *marginal* score $s(x_u, u, t) = \nabla_{x_u} \log p_t(x_u \mid u)$, which marginalises over all other frames and therefore discards all inter-frame correlations. The correct object is the *joint* score $\mathbf{S}(\mathbf{x}, t) \in \mathbb{R}^{K+1}$, whose k -th component couples x_k to all other frames through Σ_t^{-1} .

2 The AR(1) Gaussian Model

2.1 Model specification

Definition 2.1 (AR(1) Gaussian sequence). The **clean frames** $\mathbf{a} = (a_0, a_1, \dots, a_K) \in \mathbb{R}^{K+1}$ are defined by the following three ingredients, all independent of one another.

(i) **Initial distribution:**

$$a_0 \sim \mathcal{N}(\mu_0, \sigma_0^2).$$

(ii) **Transition rule (AR(1)):** for $k = 0, 1, \dots, K-1$,

$$a_{k+1} = \alpha a_k + \eta_k, \quad \eta_k \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_\eta^2), \quad \eta_k \perp a_0, \eta_k \perp \eta_j \ (j \neq k).$$

(iii) **Transition kernel:** the above implies

$$M(a_{k+1} \mid a_k) = \mathcal{N}(a_{k+1}; \alpha a_k, \sigma_\eta^2).$$

Remark 2.1 (Modelling choices). The zero-mean Gaussian noise η_k is the *maximum-entropy* distribution consistent with a given energy scale σ_η^2 , and it guarantees analytical tractability throughout: sums of Gaussians are Gaussian, and the entire trajectory $\mathbf{a}$ remains jointly Gaussian. The parameter $|\alpha| < 1$ ensures the chain does not explode, with stationary variance $\sigma_\infty^2 = \sigma_\eta^2 / (1 - \alpha^2)$. The i.i.d. assumption is on the *noise inputs* η_k , not on the frames a_k themselves, which are correlated through the recurrence.

2.2 Moments of the chain

Proposition 2.1 (Mean and variance recursions). The mean $m_k = \mathbb{E}[a_k]$ and variance $\sigma_k^2 = \text{Var}(a_k)$ satisfy the closed-form recursions:

$$m_k = \alpha^k \mu_0, \quad \sigma_k^2 = \alpha^{2k} \sigma_0^2 + \sigma_\eta^2 \frac{1 - \alpha^{2k}}{1 - \alpha^2}.$$

Proof. Mean. Apply $\mathbb{E}[\cdot]$ to $a_{k+1} = \alpha a_k + \eta_k$. By linearity and $\mathbb{E}[\eta_k] = 0$:

$$m_{k+1} = \alpha m_k.$$

Unrolling from $m_0 = \mu_0$ gives $m_k = \alpha^k \mu_0$.

Variance. Since $a_k \perp \eta_k$:

$$\sigma_{k+1}^2 = \text{Var}(\alpha a_k + \eta_k) = \alpha^2 \sigma_k^2 + \sigma_\eta^2.$$

This is a first-order linear recursion with constant forcing. Unrolling k times from σ_0^2 :

$$\sigma_k^2 = \alpha^{2k} \sigma_0^2 + \sigma_\eta^2 \sum_{j=0}^{k-1} \alpha^{2j} = \alpha^{2k} \sigma_0^2 + \sigma_\eta^2 \frac{1 - \alpha^{2k}}{1 - \alpha^2}. \quad \square$$

□

Proposition 2.2 (Covariance between frames). For $j \leq k$:

$$\text{Cov}(a_j, a_k) = \alpha^{k-j} \sigma_j^2.$$

Hence the covariance matrix $\Sigma_0 \in \mathbb{R}^{(K+1) \times (K+1)}$ has entries

$$(\Sigma_0)_{jk} = \alpha^{|j-k|} \sigma_{\min(j,k)}^2.$$

Proof. Unroll the recursion from a_j to a_k ($j \leq k$):

$$a_k = \alpha^{k-j} a_j + \sum_{i=j}^{k-1} \alpha^{k-1-i} \eta_i.$$

The noise terms η_i for $i \geq j$ are injected after time j , hence independent of a_j . Therefore:

$$\text{Cov}(a_j, a_k) = \mathbb{E} \left[\tilde{a}_j \left(\alpha^{k-j} \tilde{a}_j + \sum_{i=j}^{k-1} \alpha^{k-1-i} \eta_i \right) \right] = \alpha^{k-j} \mathbb{E}[\tilde{a}_j^2] + \sum_{i=j}^{k-1} \alpha^{k-1-i} \underbrace{\mathbb{E}[\tilde{a}_j \eta_i]}_{=0} = \alpha^{k-j} \sigma_j^2,$$

where $\tilde{a}_j = a_j - m_j$. Symmetry $\text{Cov}(a_j, a_k) = \text{Cov}(a_k, a_j)$ gives the general formula. $\square$

Remark 2.2 (Non-stationarity). Σ_0 depends on position j (not just lag $|j-k|$) because σ_j^2 varies with j . The chain becomes *stationary* (i.e. $\text{Cov}(a_j, a_k) = \alpha^{|j-k|} \sigma_\infty^2$) if and only if $\sigma_0^2 = \sigma_\infty^2 = \sigma_\eta^2 / (1 - \alpha^2)$.

3 The Clean Joint Distribution P_0

3.1 Markov factorisation

Proposition 3.1 (Markov factorisation). The joint density of the clean trajectory factorises as

$$P_0(a_0, a_1, \dots, a_K) = p_0(a_0) \prod_{k=0}^{K-1} M(a_{k+1} | a_k),$$

or explicitly:

$$P_0(\mathbf{a}) = \frac{1}{(2\pi)^{(K+1)/2} \sigma_0 \sigma_\eta^K} \exp \left(-\frac{(a_0 - \mu_0)^2}{2\sigma_0^2} - \sum_{k=0}^{K-1} \frac{(a_{k+1} - \alpha a_k)^2}{2\sigma_\eta^2} \right).$$

Proof. The chain rule of probability gives:

$$P_0(\mathbf{a}) = p(a_0) \cdot p(a_1 | a_0) \cdot p(a_2 | a_1, a_0) \cdots p(a_K | a_{K-1}, \dots, a_0).$$

The Markov property reduces each factor: $p(a_k | a_{k-1}, \dots, a_0) = M(a_k | a_{k-1})$. Substituting the Gaussian densities gives the explicit product form. $\square$ $\square$

3.2 The precision matrix

The exponent of $P_0(\mathbf{a})$ is quadratic in $\mathbf{a}$. Define the *precision matrix* Σ_0^{-1} via

$$\frac{(a_0 - \mu_0)^2}{2\sigma_0^2} + \sum_{k=0}^{K-1} \frac{(a_{k+1} - \alpha a_k)^2}{2\sigma_\eta^2} \equiv \frac{1}{2}(\mathbf{a} - \boldsymbol{\mu}_a)^\top \Sigma_0^{-1}(\mathbf{a} - \boldsymbol{\mu}_a),$$

where $\boldsymbol{\mu}_a = (\mu_0, \alpha\mu_0, \dots, \alpha^K\mu_0)^\top$.

Proposition 3.2 (Tridiagonal precision matrix). The precision matrix Σ_0^{-1} is **symmetric** and **tridiagonal** with entries:

$$\begin{aligned} (\Sigma_0^{-1})_{00} &= \frac{1}{\sigma_0^2} + \frac{\alpha^2}{\sigma_\eta^2}, \\ (\Sigma_0^{-1})_{kk} &= \frac{1 + \alpha^2}{\sigma_\eta^2}, \quad k = 1, \dots, K-1, \\ (\Sigma_0^{-1})_{KK} &= \frac{1}{\sigma_\eta^2}, \\ (\Sigma_0^{-1})_{k,k+1} &= (\Sigma_0^{-1})_{k+1,k} = -\frac{\alpha}{\sigma_\eta^2}, \quad k = 0, \dots, K-1, \\ (\Sigma_0^{-1})_{jk} &= 0, \quad |j - k| \geq 2. \end{aligned}$$

Proof. Expand the exponent and collect terms. For a symmetric matrix A , the identity

$$\mathbf{v}^\top A \mathbf{v} = \sum_i A_{ii} v_i^2 + 2 \sum_{i < j} A_{ij} v_i v_j$$

implies: A_{ii} equals the coefficient of v_i^2 , and A_{ij} ($i < j$) equals *half* the coefficient of $v_i v_j$.

Setting $\tilde{a}_k = a_k - \alpha^k \mu_0$ (centred variables):

- **Coefficient of $\tilde{a}_0^2$:** appears in $(a_0 - \mu_0)^2/\sigma_0^2$ as $1/\sigma_0^2$, and in $(a_1 - \alpha a_0)^2/\sigma_\eta^2$ as α^2/σ_η^2 . Sum: $(\Sigma_0^{-1})_{00} = 1/\sigma_0^2 + \alpha^2/\sigma_\eta^2$.
- **Coefficient of $\tilde{a}_k^2$, interior k :** frame k appears as $\tilde{a}_k^2$ in transition $k-1$ and as $\alpha^2 \tilde{a}_k^2$ in transition k . Sum: $(\Sigma_0^{-1})_{kk} = (1 + \alpha^2)/\sigma_\eta^2$.
- **Coefficient of $\tilde{a}_K^2$:** only in the last transition as $\tilde{a}_K^2/\sigma_\eta^2$. Hence $(\Sigma_0^{-1})_{KK} = 1/\sigma_\eta^2$.
- **Coefficient of $\tilde{a}_k \tilde{a}_{k+1}$:** the cross term $-2\alpha \tilde{a}_{k+1} \tilde{a}_k / \sigma_\eta^2$ in transition k gives, after halving, $(\Sigma_0^{-1})_{k,k+1} = -\alpha/\sigma_\eta^2$.
- **Terms $\tilde{a}_j \tilde{a}_k$ with $|j - k| \geq 2$:** such cross-products never appear in the expanded sum of squared residuals — a direct algebraic consequence of the Markov property. Hence $(\Sigma_0^{-1})_{jk} = 0$. $\square$

$\square$

Remark 3.1 (Conditional independence interpretation). For a jointly Gaussian vector, $(\Sigma_0^{-1})_{jk} = 0$ if and only if $a_j \perp a_k \mid \{a_\ell : \ell \neq j, k\}$. The tridiagonal structure of Σ_0^{-1} is therefore the algebraic encoding of the Markov property: conditioning on all intermediate frames screens off a_j and a_k for $|j - k| \geq 2$. The covariance matrix Σ_0 is *dense* (all entries nonzero for $\alpha \neq 0$), encoding *total* correlations including all indirect paths through intermediaries; the precision matrix encodes only *direct* couplings. More precisely, from the conditional distribution $a_j \mid \{a_k\}_{k \neq j}$, the conditional mean satisfies

$$\mathbb{E}[a_j \mid \{a_k\}_{k \neq j}] = \mu_j - \frac{1}{(\Sigma_0^{-1})_{jj}} \sum_{k \neq j} (\Sigma_0^{-1})_{jk} (a_k - \mu_k),$$

so a zero off-diagonal entry $(\Sigma_0^{-1})_{jk} = 0$ means a_k does not appear in the conditional mean of a_j given everything else.

Clean joint distribution. The AR(1) trajectory is a $(K+1)$ -dimensional Gaussian:

$$P_0(\mathbf{a}) = \frac{1}{(2\pi)^{(K+1)/2} |\Sigma_0|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{a} - \boldsymbol{\mu}_a)^\top \Sigma_0^{-1} (\mathbf{a} - \boldsymbol{\mu}_a)\right),$$

with mean $\boldsymbol{\mu}_a = (\mu_0, \alpha\mu_0, \dots, \alpha^K\mu_0)^\top$ and tridiagonal precision matrix Σ_0^{-1} given in Proposition 3.2.

4 The Noisy Joint Distribution P_t

4.1 The Ornstein-Uhlenbeck forward process

Definition 4.1 (OU diffusion per frame). At diffusion time $t \geq 0$, each clean frame a_k is independently corrupted:

$$x_k = a_k e^{-t} + \sqrt{\Delta_t} z_k, \quad z_k \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1), \quad \mathbf{z} \perp \mathbf{a}, \quad \Delta_t = 1 - e^{-2t}.$$

Note the identity $e^{-2t} + \Delta_t = 1$: signal energy plus noise energy is conserved.

The scaling rule $\sqrt{\Delta_t} z_k \sim \mathcal{N}(0, \Delta_t)$ gives:

$$x_k \mid a_k \sim \mathcal{N}(a_k e^{-t}, \Delta_t), \quad p(x_k \mid a_k) = \frac{1}{\sqrt{2\pi\Delta_t}} \exp\left(-\frac{(x_k - a_k e^{-t})^2}{2\Delta_t}\right).$$

Since the noises z_k are independent across frames, the joint conditional likelihood factorises:

$$p(\mathbf{x} \mid \mathbf{a}) = \prod_{k=0}^K \mathcal{N}(x_k; a_k e^{-t}, \Delta_t).$$

4.2 The marginal distribution of noisy observations

The noisy observations $\mathbf{x} = (x_0, \dots, x_K)$ are obtained by marginalising over the hidden clean frames:

$$P_t(\mathbf{x}) = \int_{\mathbb{R}^{K+1}} P_0(\mathbf{a}) \prod_{k=0}^K \mathcal{N}(x_k; a_k e^{-t}, \Delta_t) d\mathbf{a}. \quad (\text{Mézard's board formula})$$

We compute this integral by two independent methods.

4.3 Method 1: Linear transformation of Gaussians

Theorem 4.1 (P_t is Gaussian). The marginal $P_t(\mathbf{x})$ is a $(K+1)$ -dimensional Gaussian:

$$P_t(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_t, \Sigma_t),$$

with

$$\boldsymbol{\mu}_t = e^{-t} \boldsymbol{\mu}_a, \quad \Sigma_t = e^{-2t} \Sigma_0 + \Delta_t I_{K+1}.$$

Proof. Write the observation model in vector form:

$$\mathbf{x} = e^{-t} \mathbf{a} + \sqrt{\Delta_t} \mathbf{z}, \quad \mathbf{a} \sim \mathcal{N}(\boldsymbol{\mu}_a, \Sigma_0), \quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, I_{K+1}), \quad \mathbf{a} \perp \mathbf{z}.$$

$\mathbf{x}$ is a linear function of two independent Gaussian vectors, hence Gaussian.

Mean:

$$\boldsymbol{\mu}_t = \mathbb{E}[\mathbf{x}] = e^{-t} \mathbb{E}[\mathbf{a}] + \sqrt{\Delta_t} \mathbb{E}[\mathbf{z}] = e^{-t} \boldsymbol{\mu}_a.$$

Covariance: using independence $\mathbf{a} \perp \mathbf{z}$ and the bilinearity of covariance,

$$\Sigma_t = \text{Cov}\left(e^{-t} \mathbf{a} + \sqrt{\Delta_t} \mathbf{z}\right) = e^{-2t} \text{Cov}(\mathbf{a}) + \Delta_t \text{Cov}(\mathbf{z}) = e^{-2t} \Sigma_0 + \Delta_t I_{K+1}. \quad \square$$

□

Remark 4.1 (Physical interpretation of Σ_t). The covariance $\Sigma_t = e^{-2t} \Sigma_0 + \Delta_t I$ has two competing contributions:

- $e^{-2t} \Sigma_0$: the *signal* covariance, carrying all inter-frame AR(1) correlations, shrinking exponentially with t .
- $\Delta_t I$: the *noise* covariance, diagonal (independent noise per frame), growing from 0 to 1 as t increases.

At $t = 0$: $\Sigma_0 = \Sigma_0$ (pure signal). As $t \rightarrow \infty$: $\Sigma_t \rightarrow I$ (pure noise, all correlation erased).

4.4 Method 2: Direct Gaussian integral

We now verify Theorem 4.1 by explicitly computing the integral, which is instructive in its own right.

Proof of Theorem 4.1 by direct integration. We compute $P_t(\mathbf{x}) = \int_{\mathbb{R}^{K+1}} P_0(\mathbf{a}) p(\mathbf{x} | \mathbf{a}) d\mathbf{a}$ by treating the integrand as a Gaussian in $\mathbf{a}$ and completing the square.

Step 1: combine the two exponentials. Up to positive normalisation constants (absorbed into the proportionality),

$$P_0(\mathbf{a}) \cdot p(\mathbf{x} | \mathbf{a}) \propto \exp\left(-\frac{1}{2}(\mathbf{a} - \boldsymbol{\mu}_a)^\top \Sigma_0^{-1}(\mathbf{a} - \boldsymbol{\mu}_a) - \frac{1}{2\Delta_t}\|\mathbf{x} - e^{-t}\mathbf{a}\|^2\right).$$

Expand the second quadratic (writing $\|\mathbf{x} - e^{-t}\mathbf{a}\|^2 = \mathbf{x}^\top \mathbf{x} - 2e^{-t}\mathbf{x}^\top \mathbf{a} + e^{-2t}\mathbf{a}^\top \mathbf{a}$):

$$\frac{1}{2\Delta_t}\|\mathbf{x} - e^{-t}\mathbf{a}\|^2 = \frac{e^{-2t}}{2\Delta_t}\mathbf{a}^\top \mathbf{a} - \frac{e^{-t}}{\Delta_t}\mathbf{x}^\top \mathbf{a} + \frac{1}{2\Delta_t}\mathbf{x}^\top \mathbf{x}.$$

The last term does not depend on $\mathbf{a}$; absorb it into the proportionality. Expand the first quadratic similarly:

$$\frac{1}{2}(\mathbf{a} - \boldsymbol{\mu}_a)^\top \Sigma_0^{-1}(\mathbf{a} - \boldsymbol{\mu}_a) = \frac{1}{2}\mathbf{a}^\top \Sigma_0^{-1}\mathbf{a} - \mathbf{a}^\top \Sigma_0^{-1}\boldsymbol{\mu}_a + \frac{1}{2}\boldsymbol{\mu}_a^\top \Sigma_0^{-1}\boldsymbol{\mu}_a.$$

The last term is a constant in $\mathbf{a}$; absorb it. Collecting all $\mathbf{a}$ -dependent terms in the exponent:

$$-\frac{1}{2}\mathbf{a}^\top \underbrace{\left(\Sigma_0^{-1} + \frac{e^{-2t}}{\Delta_t}I\right)}_{\boldsymbol{\Lambda}} \mathbf{a} + \mathbf{a}^\top \underbrace{\left(\Sigma_0^{-1}\boldsymbol{\mu}_a + \frac{e^{-t}}{\Delta_t}\mathbf{x}\right)}_{\mathbf{h}}.$$

Step 2: complete the square in $\mathbf{a}$. Define

$$\begin{aligned}\boldsymbol{\Lambda} &= \Sigma_0^{-1} + \frac{e^{-2t}}{\Delta_t}I \in \mathbb{R}^{(K+1) \times (K+1)}, \\ \mathbf{h} &= \Sigma_0^{-1}\boldsymbol{\mu}_a + \frac{e^{-t}}{\Delta_t}\mathbf{x}, \\ \mathbf{m} &= \boldsymbol{\Lambda}^{-1}\mathbf{h} \quad (\text{posterior mean of } \mathbf{a} \text{ given } \mathbf{x}).\end{aligned}$$

The standard identity $-\frac{1}{2}\mathbf{a}^\top \boldsymbol{\Lambda} \mathbf{a} + \mathbf{a}^\top \mathbf{h} = -\frac{1}{2}(\mathbf{a} - \mathbf{m})^\top \boldsymbol{\Lambda}(\mathbf{a} - \mathbf{m}) + \frac{1}{2}\mathbf{m}^\top \boldsymbol{\Lambda} \mathbf{m}$ rewrites the exponent as a Gaussian in $\mathbf{a}$ centred at $\mathbf{m}$, plus a term $\frac{1}{2}\mathbf{m}^\top \boldsymbol{\Lambda} \mathbf{m}$ that depends only on $\mathbf{x}$.

Step 3: perform the Gaussian integral over $\mathbf{a}$.

$$\int_{\mathbb{R}^{K+1}} \exp\left(-\frac{1}{2}(\mathbf{a} - \mathbf{m})^\top \boldsymbol{\Lambda}(\mathbf{a} - \mathbf{m})\right) d\mathbf{a} = (2\pi)^{(K+1)/2} |\boldsymbol{\Lambda}|^{-1/2}.$$

The remaining $\mathbf{x}$ -dependent part of the exponent is:

$$Q(\mathbf{x}) \equiv \frac{1}{2}\mathbf{m}^\top \boldsymbol{\Lambda} \mathbf{m} - \frac{1}{2\Delta_t}\mathbf{x}^\top \mathbf{x} = \frac{1}{2}\mathbf{h}^\top \boldsymbol{\Lambda}^{-1}\mathbf{h} - \frac{1}{2\Delta_t}\mathbf{x}^\top \mathbf{x}.$$

Step 4: identify $Q(\mathbf{x})$ as a Gaussian quadratic in $\mathbf{x}$. Substitute $\mathbf{h} = \Sigma_0^{-1}\boldsymbol{\mu}_a + \frac{e^{-t}}{\Delta_t}\mathbf{x}$ and apply the **Woodbury identity** $\boldsymbol{\Lambda}^{-1} = \left(\Sigma_0^{-1} + \frac{e^{-2t}}{\Delta_t}I\right)^{-1} = \Sigma_0 - e^{-2t}\Sigma_0\Sigma_t^{-1}\Sigma_0$ (derivable from $(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}$ with $A = \Sigma_0^{-1}$, $U = V^\top = I$, $C = \frac{e^{-2t}}{\Delta_t}I$). After expanding and collecting the $\mathbf{x}^\top(\cdot)\mathbf{x}$ terms, one finds that all cross-terms reduce to:

$$Q(\mathbf{x}) = -\frac{1}{2}(\mathbf{x} - e^{-t}\boldsymbol{\mu}_a)^\top \Sigma_t^{-1}(\mathbf{x} - e^{-t}\boldsymbol{\mu}_a) + \text{const},$$

where $\Sigma_t = e^{-2t}\Sigma_0 + \Delta_t I$ (one verifies: $\frac{1}{\Delta_t}I - \frac{e^{-2t}}{\Delta_t^2}\Sigma_0\Lambda^{-1} = \Sigma_t^{-1}$ by multiplying both sides by Σ_t and using $\Lambda = \Sigma_0^{-1} + \frac{e^{-2t}}{\Delta_t}I$).

Conclusion. Assembling all factors, $P_t(\mathbf{x}) \propto \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_t)^\top \Sigma_t^{-1}(\mathbf{x} - \boldsymbol{\mu}_t)\right)$ with $\boldsymbol{\mu}_t = e^{-t}\boldsymbol{\mu}_a$ and $\Sigma_t = e^{-2t}\Sigma_0 + \Delta_t I$, confirming it is $\mathcal{N}(\boldsymbol{\mu}_t, \Sigma_t)$. $\square$

Noisy joint distribution.

$$P_t(\mathbf{x}) = \frac{1}{(2\pi)^{(K+1)/2}|\Sigma_t|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_t)^\top \Sigma_t^{-1}(\mathbf{x} - \boldsymbol{\mu}_t)\right),$$

with $\boldsymbol{\mu}_t = e^{-t}\boldsymbol{\mu}_a$ and $\Sigma_t = e^{-2t}\Sigma_0 + \Delta_t I_{K+1}$.

5 The Joint Score Function

5.1 Derivation

Definition 5.1 (Joint score). The **joint score** at diffusion time t is:

$$\mathbf{S}(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log P_t(\mathbf{x}) \in \mathbb{R}^{K+1}.$$

Its k -th component $S_k(\mathbf{x}, t) = \partial_{x_k} \log P_t(\mathbf{x})$ measures how increasing x_k changes the log-probability of the entire noisy trajectory.

Theorem 5.1 (Joint score is linear).

$$\mathbf{S}(\mathbf{x}, t) = -\Sigma_t^{-1}(\mathbf{x} - \boldsymbol{\mu}_t).$$

Proof. Since $P_t(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_t, \Sigma_t)$, its log-density is:

$$\log P_t(\mathbf{x}) = -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_t)^\top \Sigma_t^{-1}(\mathbf{x} - \boldsymbol{\mu}_t) - \underbrace{\frac{K+1}{2} \log(2\pi) - \frac{1}{2} \log |\Sigma_t|}_{\text{constant in } \mathbf{x}}.$$

It therefore suffices to differentiate $-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_t)^\top \Sigma_t^{-1}(\mathbf{x} - \boldsymbol{\mu}_t)$ with respect to $\mathbf{x}$. Expand the quadratic form using bilinearity:

$$(\mathbf{x} - \boldsymbol{\mu}_t)^\top \Sigma_t^{-1}(\mathbf{x} - \boldsymbol{\mu}_t) = \mathbf{x}^\top \Sigma_t^{-1} \mathbf{x} - 2 \boldsymbol{\mu}_t^\top \Sigma_t^{-1} \mathbf{x} + \underbrace{\boldsymbol{\mu}_t^\top \Sigma_t^{-1} \boldsymbol{\mu}_t}_{\text{constant in } \mathbf{x}}.$$

We differentiate the two remaining $\mathbf{x}$ -dependent terms separately.

Step 1: gradient of the quadratic term. For any symmetric matrix $A \in \mathbb{R}^{n \times n}$ and $\mathbf{v} \in \mathbb{R}^n$, compute the m -th partial derivative of $\mathbf{v}^\top A \mathbf{v} = \sum_{i,j} A_{ij} v_i v_j$ by the product rule:

$$\frac{\partial}{\partial v_m} (\mathbf{v}^\top A \mathbf{v}) = \sum_j A_{mj} v_j + \sum_i A_{im} v_i = (A \mathbf{v})_m + (A^\top \mathbf{v})_m.$$

Since Σ_t^{-1} is symmetric ($\Sigma_t^{-1} = (\Sigma_t^{-1})^\top$, as the inverse of a symmetric matrix is symmetric), the two sums are equal and:

$$\nabla_{\mathbf{x}}(\mathbf{x}^\top \Sigma_t^{-1} \mathbf{x}) = 2 \Sigma_t^{-1} \mathbf{x}.$$

Step 2: gradient of the linear term. The term $-2 \boldsymbol{\mu}_t^\top \Sigma_t^{-1} \mathbf{x}$ is linear in $\mathbf{x}$. Write it as $\mathbf{c}^\top \mathbf{x}$ where $\mathbf{c} = -2 \Sigma_t^{-1} \boldsymbol{\mu}_t$ is a constant vector (independent of $\mathbf{x}$). Since $\nabla_{\mathbf{x}}(\mathbf{c}^\top \mathbf{x}) = \mathbf{c}$ for any constant $\mathbf{c}$:

$$\nabla_{\mathbf{x}}(-2 \boldsymbol{\mu}_t^\top \Sigma_t^{-1} \mathbf{x}) = -2 \Sigma_t^{-1} \boldsymbol{\mu}_t.$$

Combining Steps 1 and 2:

$$\begin{aligned} \mathbf{S}(\mathbf{x}, t) &= \nabla_{\mathbf{x}}[\log P_t(\mathbf{x})] \\ &= -\frac{1}{2} \nabla_{\mathbf{x}}(\mathbf{x}^\top \Sigma_t^{-1} \mathbf{x} - 2 \boldsymbol{\mu}_t^\top \Sigma_t^{-1} \mathbf{x}) \\ &= -\frac{1}{2} (2 \Sigma_t^{-1} \mathbf{x} - 2 \Sigma_t^{-1} \boldsymbol{\mu}_t) \\ &= -\Sigma_t^{-1} (\mathbf{x} - \boldsymbol{\mu}_t). \end{aligned}$$

□

Remark 5.1 (Linearity is a Gaussian property). The score is affine in $\mathbf{x}$:

$$\mathbf{S}(\mathbf{x}, t) = \underbrace{-\Sigma_t^{-1}}_{\mathbf{A}(t)} \mathbf{x} + \underbrace{\Sigma_t^{-1} \boldsymbol{\mu}_t}_{\mathbf{b}(t)}.$$

This linearity is specific to Gaussian distributions. For a non-Gaussian prior P_0 (e.g. a mixture of Gaussians), P_t is non-Gaussian for small t and the score is a nonlinear function of $\mathbf{x}$. The AR(1) Gaussian model is the unique tractable case where linearity holds exactly at all t .

5.2 The k -th component and inter-frame coupling

The k -th component of the joint score is:

Corollary 5.1 (Component-wise joint score).

$$S_k(\mathbf{x}, t) = - \sum_{j=0}^K (\Sigma_t^{-1})_{kj} (x_j - \mu_{t,j}),$$

where $\mu_{t,j} = e^{-t} \alpha^j \mu_0$. This couples x_k to **all other frames** x_j through the precision matrix entries $(\Sigma_t^{-1})_{kj}$.

Remark 5.2 (Limiting regimes).

- $t \rightarrow 0$ (**clean signal**): $\Sigma_t^{-1} \rightarrow \Sigma_0^{-1}$, which is tridiagonal. Hence S_k depends only on x_{k-1} , x_k , x_{k+1} — nearest-neighbour coupling only.
- $t \rightarrow \infty$ (**pure noise**): $\Sigma_t^{-1} \rightarrow \Delta_t^{-1} I \rightarrow I$, so $S_k \approx -x_k / \Delta_t$ — each frame's score depends only on itself, all inter-frame coupling vanishes.
- **Intermediate t** : Σ_t^{-1} is not tridiagonal in general. Adding $\Delta_t I$ to Σ_0 and inverting fills in the structural zeros of Σ_0^{-1} , producing small but nonzero off-tridiagonal entries. The tridiagonal structure is only approximate for $t > 0$.

6 Connection to Mézard's Formula

The formula derived in Theorem 5.1 has an equivalent probabilistic representation as a conditional expectation, which is what Mézard wrote on the board during the meeting.

Theorem 6.1 (Score as a conditional expectation). The k -th component of the joint score satisfies:

$$S_k(\mathbf{x}, t) = \left\langle \frac{a_k e^{-t} - x_k}{\Delta_t} \right\rangle_{\mathbf{a}|\mathbf{x}},$$

where $\langle \cdot \rangle_{\mathbf{a}|\mathbf{x}}$ denotes expectation under the posterior $P(\mathbf{a} | \mathbf{x}) \propto P_0(\mathbf{a}) \prod_j \mathcal{N}(x_j; a_j e^{-t}, \Delta_t)$.

Proof. By definition:

$$S_k(\mathbf{x}, t) = \partial_{x_k} \log P_t(\mathbf{x}) = \frac{\partial_{x_k} P_t(\mathbf{x})}{P_t(\mathbf{x})}.$$

Write $P_t(\mathbf{x}) = \int P_0(\mathbf{a}) \prod_j \mathcal{N}(x_j; a_j e^{-t}, \Delta_t) d\mathbf{a}$. Only the k -th factor depends on x_k . Using the identity $\partial_x \log \mathcal{N}(x; \mu, \sigma^2) = (\mu - x)/\sigma^2$:

$$\partial_{x_k} \mathcal{N}(x_k; a_k e^{-t}, \Delta_t) = \mathcal{N}(x_k; a_k e^{-t}, \Delta_t) \cdot \frac{a_k e^{-t} - x_k}{\Delta_t}.$$

Therefore:

$$\partial_{x_k} P_t(\mathbf{x}) = \int P_0(\mathbf{a}) \prod_j \mathcal{N}(x_j; a_j e^{-t}, \Delta_t) \cdot \frac{a_k e^{-t} - x_k}{\Delta_t} d\mathbf{a}.$$

Dividing by $P_t(\mathbf{x})$ turns the integral into an expectation under the posterior $P(\mathbf{a} | \mathbf{x})$:

$$S_k(\mathbf{x}, t) = \left\langle \frac{a_k e^{-t} - x_k}{\Delta_t} \right\rangle_{\mathbf{a}|\mathbf{x}}. \quad \square$$

□

Remark 6.1 (Denoising interpretation). The formula $S_k = (e^{-t} \langle a_k \rangle_{\mathbf{a}|\mathbf{x}} - x_k) / \Delta_t$ is a **denoising residual**: the expected clean signal scaled back to the observed level, minus the actual noisy observation, divided by the noise variance. Three features deserve emphasis:

1. The expectation $\langle a_k \rangle_{\mathbf{a}|\mathbf{x}}$ conditions on the **full noisy trajectory** $\mathbf{x}$, not just x_k . This is the key difference from the marginal score.
2. For Gaussian AR(1), the posterior $P(\mathbf{a} | \mathbf{x})$ is Gaussian, and the posterior mean $\langle \mathbf{a} \rangle_{\mathbf{a}|\mathbf{x}}$ is computable exactly by the **Kalman smoother**.
3. The posterior mean is **linear** in $\mathbf{x}$: $\langle \mathbf{a} \rangle_{\mathbf{a}|\mathbf{x}} = e^{-t} \Sigma_0 \Sigma_t^{-1} (\mathbf{x} - \boldsymbol{\mu}_t) + \boldsymbol{\mu}_a$, which is consistent with the explicit formula $S_k = -[\Sigma_t^{-1} (\mathbf{x} - \boldsymbol{\mu}_t)]_k$.

6.1 Equivalence of the two formulas

We now prove algebraically that Mézard's conditional expectation formula and the explicit linear formula of Theorem 5.1 are identical. The key ingredient is the **posterior mean** of the clean frames.

Lemma 6.1 (Gaussian posterior mean). For Gaussian $P_0(\mathbf{a}) = \mathcal{N}(\mathbf{a}; \boldsymbol{\mu}_a, \Sigma_0)$ and Gaussian likelihood $p(\mathbf{x} | \mathbf{a}) = \prod_k \mathcal{N}(x_k; a_k e^{-t}, \Delta_t)$, the posterior $P(\mathbf{a} | \mathbf{x})$ is Gaussian with mean:

$$\langle \mathbf{a} \rangle_{\mathbf{a}|\mathbf{x}} = \boldsymbol{\mu}_a + e^{-t} \Sigma_0 \Sigma_t^{-1} (\mathbf{x} - \boldsymbol{\mu}_t).$$

Proof. The posterior precision is $\Lambda = \Sigma_0^{-1} + \frac{e^{-2t}}{\Delta_t} I$ (from Step 2 of the direct integral proof) and the posterior mean is $\mathbf{m} = \Lambda^{-1}(\Sigma_0^{-1} \boldsymbol{\mu}_a + \frac{e^{-t}}{\Delta_t} \mathbf{x})$. Apply the Woodbury identity $\Lambda^{-1} = \Sigma_0 - e^{-2t} \Sigma_0 \Sigma_t^{-1} \Sigma_0$ and the relation $\frac{e^{-2t}}{\Delta_t} \Sigma_0 = \Sigma_t^{-1} \Sigma_0 \cdot \frac{e^{-2t} \Sigma_0}{\Delta_t} \dots$

More directly: use the standard Gaussian conditioning formula. Because $\mathbf{x} = e^{-t} \mathbf{a} + \sqrt{\Delta_t} \mathbf{z}$ with $\mathbf{a} \perp \mathbf{z}$, the joint covariance between $\mathbf{a}$ and $\mathbf{x}$ is $\text{Cov}(\mathbf{a}, \mathbf{x}) = e^{-t} \text{Cov}(\mathbf{a}, \mathbf{a}) = e^{-t} \Sigma_0$. The conditional mean formula for jointly Gaussian variables gives:

$$\langle \mathbf{a} \rangle_{\mathbf{a}|\mathbf{x}} = \boldsymbol{\mu}_a + \text{Cov}(\mathbf{a}, \mathbf{x}) [\text{Cov}(\mathbf{x}, \mathbf{x})]^{-1} (\mathbf{x} - \boldsymbol{\mu}_t) = \boldsymbol{\mu}_a + e^{-t} \Sigma_0 \Sigma_t^{-1} (\mathbf{x} - \boldsymbol{\mu}_t). \quad \square$$

□

Proposition 6.1 (Equivalence of the two score formulas). The Mézard formula and the linear formula agree exactly:

$$\left\langle \frac{a_k e^{-t} - x_k}{\Delta_t} \right\rangle_{\mathbf{a}|\mathbf{x}} = -[\Sigma_t^{-1} (\mathbf{x} - \boldsymbol{\mu}_t)]_k.$$

Proof. Since x_k is a constant under $\langle \cdot \rangle_{\mathbf{a}|\mathbf{x}}$:

$$\left\langle \frac{a_k e^{-t} - x_k}{\Delta_t} \right\rangle_{\mathbf{a}|\mathbf{x}} = \frac{e^{-t} \langle a_k \rangle_{\mathbf{a}|\mathbf{x}} - x_k}{\Delta_t}.$$

Substitute Lemma 6.1 for the k -th component:

$$= \frac{e^{-t} (\mu_{a,k} + e^{-t} [\Sigma_0 \Sigma_t^{-1} (\mathbf{x} - \boldsymbol{\mu}_t)]_k) - x_k}{\Delta_t}.$$

Use $e^{-t} \mu_{a,k} = \mu_{t,k}$ (since $\boldsymbol{\mu}_t = e^{-t} \boldsymbol{\mu}_a$) and write $e^{-2t} \Sigma_0 = \Sigma_t - \Delta_t I$ (rearranging $\Sigma_t = e^{-2t} \Sigma_0 + \Delta_t I$), so $e^{-2t} \Sigma_0 \Sigma_t^{-1} = (\Sigma_t - \Delta_t I) \Sigma_t^{-1} = I - \Delta_t \Sigma_t^{-1}$:

$$= \frac{\mu_{t,k} + [I - \Delta_t \Sigma_t^{-1}]_k (\mathbf{x} - \boldsymbol{\mu}_t) - x_k}{\Delta_t}.$$

Expanding $[I - \Delta_t \Sigma_t^{-1}]_k (\mathbf{x} - \boldsymbol{\mu}_t) = (x_k - \mu_{t,k}) - \Delta_t [\Sigma_t^{-1} (\mathbf{x} - \boldsymbol{\mu}_t)]_k$:

$$= \frac{\mu_{t,k} + (x_k - \mu_{t,k}) - \Delta_t [\Sigma_t^{-1} (\mathbf{x} - \boldsymbol{\mu}_t)]_k - x_k}{\Delta_t} = \frac{-\Delta_t [\Sigma_t^{-1} (\mathbf{x} - \boldsymbol{\mu}_t)]_k}{\Delta_t} = -[\Sigma_t^{-1} (\mathbf{x} - \boldsymbol{\mu}_t)]_k. \quad \square$$

□

7 Summary and Notation Table

7.1 The complete derivation chain

$$\underbrace{P_0(\mathbf{a})}_{\text{AR(1) prior}} \xrightarrow{\text{OU noise, independent per frame}} \underbrace{P_t(\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_t, \Sigma_t)}_{\text{noisy joint dist.}} \xrightarrow{\nabla_{\mathbf{x}} \log} \underbrace{\mathbf{S}(\mathbf{x}, t) = -\Sigma_t^{-1}(\mathbf{x} - \boldsymbol{\mu}_t)}_{\text{joint score}}$$

7.2 Key formulas collected

Object	Formula	Key property
Markov prior	$P_0(\mathbf{a}) = p_0(a_0) \prod_k M(a_{k+1} \mid a_k)$	Markov factorisation
Covariance	$(\Sigma_0)_{jk} = \alpha^{ j-k } \sigma_{\min(j,k)}^2$	Dense
Precision	$(\Sigma_0^{-1})_{k,k\pm 1} = -\alpha / \sigma_\eta^2$	Tridiagonal (sparse)
OU variance	$\Delta_t = 1 - e^{-2t}$	$e^{-2t} + \Delta_t = 1$
Noisy covariance	$\Sigma_t = e^{-2t} \Sigma_0 + \Delta_t I$	Signal + noise
Noisy mean	$\boldsymbol{\mu}_t = e^{-t} \boldsymbol{\mu}_a$	Shrinks by e^{-t}
Joint score (algebraic)	$\mathbf{S} = -\Sigma_t^{-1}(\mathbf{x} - \boldsymbol{\mu}_t)$	Linear in $\mathbf{x}$
Joint score (Mézard)	$S_k = \langle (a_k e^{-t} - x_k) / \Delta_t \rangle_{\mathbf{a} \mathbf{x}}$	Denoising residual
Marginal score (old)	$s(x_k, k, t) = -(x_k - \mu_{t,k}) / T_{kk}$	Ignores other frames

7.3 Notation reference

Symbol	Meaning
$k = 0, \dots, K$	Internal time index (frame number), $K + 1$ frames total
$t \geq 0$	Diffusion time (OU noise level)
$a_k \in \mathbb{R}$	Clean frame at internal time k
$x_k \in \mathbb{R}$	Noisy observation of frame k at diffusion time t
$\mathbf{a}, \mathbf{x} \in \mathbb{R}^{K+1}$	Full clean / noisy trajectory
$\alpha \in (-1, 1)$	AR(1) contraction coefficient
σ_0^2	Variance of initial frame a_0
σ_η^2	AR(1) noise variance (variance of each η_k)
$\sigma_\infty^2 = \sigma_\eta^2 / (1 - \alpha^2)$	Stationary variance
$\Delta_t = 1 - e^{-2t}$	OU noise variance at time t
Σ_0	$(K + 1) \times (K + 1)$ covariance of clean chain
$\Sigma_t = e^{-2t}\Sigma_0 + \Delta_t I$	Covariance of noisy observations
$\boldsymbol{\mu}_a = (\mu_0, \alpha\mu_0, \dots, \alpha^K\mu_0)^\top$	Mean of clean chain
$\boldsymbol{\mu}_t = e^{-t}\boldsymbol{\mu}_a$	Mean of noisy observations
$M(a_{k+1} \mid a_k)$	AR(1) transition kernel
$P_0(\mathbf{a})$	Joint distribution of clean trajectory
$P_t(\mathbf{x})$	Joint distribution of noisy trajectory
$\mathbf{S}(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log P_t(\mathbf{x})$	Joint score (correct object)
$s(x_k, k, t) = \nabla_{x_k} \log p_t(x_k \mid k)$	Marginal score (old, incorrect for propagator)

8 Discussion: Alternative Noise Models for Sequences

8.1 What does it mean to apply diffusion noise to a sequence?

In our derivation we applied the OU process *independently and identically* to each frame:

$$x_k = a_k e^{-t} + \sqrt{\Delta_t} z_k, \quad z_k \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1), \quad k = 0, \dots, K.$$

This choice is natural but is not the only possibility. The choice of noise model fundamentally changes $P_t(\mathbf{x})$, the precision matrix Σ_t^{-1} , and therefore the structure of the joint score. Below we compare three alternatives systematically.

8.2 Case 1: i.i.d. OU per frame (our setting)

The noise covariance is $\text{Cov}(\mathbf{z}) = I_{K+1}$, so:

$$\Sigma_t = e^{-2t}\Sigma_0 + \Delta_t I_{K+1}.$$

The noise term $\Delta_t I$ is *diagonal* — it adds no coupling between frames. All inter-frame structure in Σ_t comes entirely from the signal term $e^{-2t}\Sigma_0$, which decays to zero as $t \rightarrow \infty$. This is what makes the score coupling vanish at large t .

The joint distribution $P_t(\mathbf{x})$ is Gaussian for all $t \geq 0$, regardless of whether P_0 is Gaussian. This is because convolving any distribution with a Gaussian (the OU noise) yields a distribution that is at least as smooth, and for Gaussian P_0 it remains Gaussian exactly.

8.3 Case 2: correlated OU noise across frames

Suppose instead the noise has a covariance structure $C \neq I$:

$$\mathbf{x} = e^{-t}\mathbf{a} + \sqrt{\Delta_t} C^{1/2} \mathbf{z}, \quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, I_{K+1}).$$

Then $\text{Cov}(\sqrt{\Delta_t} C^{1/2} \mathbf{z}) = \Delta_t C$, and:

$$\Sigma_t^{(C)} = e^{-2t}\Sigma_0 + \Delta_t C.$$

If P_0 is Gaussian, P_t remains Gaussian — the closure property holds. However, the precision matrix becomes $(\Sigma_t^{(C)})^{-1}$, which has a different sparsity structure than in Case 1. In particular, if C is itself dense (e.g. correlated noise between consecutive frames), the noise term *adds* off-diagonal entries, potentially strengthening rather than erasing inter-frame coupling in the score. This would be physically meaningful if the noise represents a correlated perturbation (e.g. camera shake affecting all frames simultaneously).

Remark 8.1. A natural choice for a sequence is a *temporally correlated* noise with covariance $C_{jk} = \rho^{|j-k|}$ for some $\rho \in (0, 1)$. This adds a second tridiagonal structure to Σ_t , and the precision $(\Sigma_t^{(C)})^{-1}$ remains banded but with potentially wider bandwidth.

8.4 Case 3: a single shared noise applied to the full sequence

At the opposite extreme, one could apply a single scalar noise $\xi \sim \mathcal{N}(0, 1)$ to the entire trajectory:

$$x_k = a_k e^{-t} + \sqrt{\Delta_t} \xi \quad \text{for all } k.$$

Then $\text{Cov}(\mathbf{x} \mid \mathbf{a}) = \Delta_t \mathbf{1}\mathbf{1}^\top$ (a rank-1 matrix, where $\mathbf{1} = (1, \dots, 1)^\top$). This is a *highly degenerate* noise: all frames receive exactly the same random perturbation. Even for Gaussian P_0 , the resulting $P_t(\mathbf{x})$ is Gaussian, but:

$$\Sigma_t^{(\text{shared})} = e^{-2t}\Sigma_0 + \Delta_t \mathbf{1}\mathbf{1}^\top,$$

which is rank- $(K + 1) + 1$ in general but structurally very different from Case 1. The noise now *introduces* inter-frame correlation rather than destroying it, so the score at large t does not simplify to $-\mathbf{x}/\Delta_t$ — instead it retains a non-trivial coupling through the rank-1 noise term.

8.5 What if P_0 is not Gaussian?

If $P_0(\mathbf{a})$ is not Gaussian (e.g. a mixture of Gaussians, or a distribution with heavy tails), the answer depends on the noise model:

- **i.i.d. OU (Case 1):** $P_t(\mathbf{x})$ is the convolution of P_0 with a Gaussian kernel. For small t this preserves the non-Gaussian structure of P_0 ; for large t the Gaussian kernel dominates and $P_t \approx \mathcal{N}(\mathbf{0}, I)$. At intermediate t the score $\mathbf{S}(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log P_t(\mathbf{x})$ is *nonlinear* in $\mathbf{x}$ — the linear formula $-\Sigma_t^{-1}(\mathbf{x} - \boldsymbol{\mu}_t)$ no longer applies.
- **Mixture of Gaussians:** if $P_0(\mathbf{a}) = \sum_c \pi_c \mathcal{N}(\mathbf{a}; \boldsymbol{\mu}_c, \Sigma_c)$, then $P_t(\mathbf{x}) = \sum_c \pi_c \mathcal{N}(\mathbf{x}; e^{-t}\boldsymbol{\mu}_c, e^{-2t}\Sigma_c)$ — still a mixture of Gaussians. The score is a *weighted average* of the per-component scores, with weights depending on the posterior component probabilities given $\mathbf{x}$. This is nonlinear in $\mathbf{x}$ in general.

Key takeaway. The Gaussian AR(1) model with i.i.d. OU noise is the *unique* setting in this family where: (i) P_t is Gaussian for all t , (ii) the joint score is linear in $\mathbf{x}$, (iii) the score has a closed-form precision structure, and (iv) the Kalman smoother provides the exact posterior. Any departure from Gaussianity in P_0 breaks (i)–(iv).

9 Extension to Multivariate Frames

9.1 Setup

Suppose each frame is now a d -dimensional vector:

$$\mathbf{a}_k \in \mathbb{R}^d, \quad k = 0, 1, \dots, K.$$

The full trajectory is $\mathbf{a} = (\mathbf{a}_0^\top, \mathbf{a}_1^\top, \dots, \mathbf{a}_K^\top)^\top \in \mathbb{R}^{d(K+1)}$. The AR(1) dynamics become:

$$\mathbf{a}_{k+1} = A \mathbf{a}_k + \boldsymbol{\eta}_k, \quad \boldsymbol{\eta}_k \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mathbf{0}, Q),$$

where $A \in \mathbb{R}^{d \times d}$ is the transition matrix (replacing the scalar α) and $Q \in \mathbb{R}^{d \times d}$ is the noise covariance matrix (replacing σ_η^2).

The initial distribution is $\mathbf{a}_0 \sim \mathcal{N}(\boldsymbol{\mu}_0, \Pi_0)$ for some $\boldsymbol{\mu}_0 \in \mathbb{R}^d$ and positive definite $\Pi_0 \in \mathbb{R}^{d \times d}$.

9.2 Does $P_0(\mathbf{a})$ remain Gaussian?

Proposition 9.1. For multivariate $\mathbf{a}_k \in \mathbb{R}^d$ with the AR(1) dynamics above, the joint distribution $P_0(\mathbf{a}) = P_0(\mathbf{a}_0, \dots, \mathbf{a}_K)$ is a $d(K+1)$ -dimensional Gaussian.

Proof. The argument is identical to the scalar case. The Markov factorisation holds:

$$P_0(\mathbf{a}_0, \dots, \mathbf{a}_K) = p_0(\mathbf{a}_0) \prod_{k=0}^{K-1} M(\mathbf{a}_{k+1} \mid \mathbf{a}_k),$$

where $M(\mathbf{a}_{k+1} \mid \mathbf{a}_k) = \mathcal{N}(\mathbf{a}_{k+1}; A\mathbf{a}_k, Q)$. Each factor is Gaussian in the joint variable $(\mathbf{a}_k, \mathbf{a}_{k+1})$, and the product of Gaussians with a shared linear structure is Gaussian. $\square$

9.3 The covariance matrix in the multivariate case

The mean and covariance of $\mathbf{a}_k$ generalise directly:

$$\boldsymbol{\mu}_k = A^k \boldsymbol{\mu}_0, \quad \Pi_k = A^k \Pi_0 (A^k)^\top + \sum_{j=0}^{k-1} A^j Q (A^j)^\top.$$

The cross-covariance between frames $j \leq k$ is:

$$\text{Cov}(\mathbf{a}_j, \mathbf{a}_k) = A^{k-j} \Pi_j \in \mathbb{R}^{d \times d}.$$

The full joint covariance $\Sigma_0 \in \mathbb{R}^{d(K+1) \times d(K+1)}$ is now a *block matrix* with $d \times d$ blocks:

$$(\Sigma_0)_{jk} = A^{|j-k|} \Pi_{\min(j,k)} \in \mathbb{R}^{d \times d}.$$

9.4 The precision matrix: block-tridiagonal structure

Proposition 9.2 (Block-tridiagonal precision). The precision matrix Σ_0^{-1} in the multivariate AR(1) case is **block-tridiagonal**, with $d \times d$ blocks:

$$\begin{aligned} (\Sigma_0^{-1})_{00} &= \Pi_0^{-1} + A^\top Q^{-1} A, \\ (\Sigma_0^{-1})_{kk} &= Q^{-1} + A^\top Q^{-1} A, \quad k = 1, \dots, K-1, \\ (\Sigma_0^{-1})_{KK} &= Q^{-1}, \\ (\Sigma_0^{-1})_{k,k+1} &= [(\Sigma_0^{-1})_{k+1,k}]^\top = -Q^{-1} A, \quad k = 0, \dots, K-1. \end{aligned}$$

The derivation is the same as the scalar case: expand the exponent, read off block coefficients, and use the symmetry of the full matrix. The scalar entries $1/\sigma_0^2$, $(1 + \alpha^2)/\sigma_\eta^2$, $-\alpha/\sigma_\eta^2$ are simply replaced by the $d \times d$ matrices Π_0^{-1} , $Q^{-1} + A^\top Q^{-1} A$, $-Q^{-1} A$.

9.5 The OU diffusion and joint score in the multivariate case

Applying i.i.d. OU noise independently to each frame component:

$$\mathbf{x}_k = e^{-t} \mathbf{a}_k + \sqrt{\Delta_t} \mathbf{z}_k, \quad \mathbf{z}_k \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mathbf{0}, I_d).$$

The result carries through exactly:

$$P_t(\mathbf{x}) = \mathcal{N}(\mathbf{x}; e^{-t} \boldsymbol{\mu}_a, e^{-2t} \Sigma_0 + \Delta_t I_{d(K+1)}),$$

and the joint score is:

$$\mathbf{S}(\mathbf{x}, t) = -\Sigma_t^{-1}(\mathbf{x} - e^{-t} \boldsymbol{\mu}_a) \in \mathbb{R}^{d(K+1)}.$$

The k -th *block* of the score (a d -vector) is:

$$\mathbf{S}_k(\mathbf{x}, t) = -\sum_{j=0}^K (\Sigma_t^{-1})_{kj} (\mathbf{x}_j - e^{-t} \boldsymbol{\mu}_j) \in \mathbb{R}^d.$$

Everything remains Gaussian, the score remains linear, and the precision matrix remains block-tridiagonal (approximately, for $t > 0$). The Kalman smoother generalises to the vector case without modification.

Remark 9.1 (What changes). The main additional complication is that A is now a matrix, which may not commute with Q . The off-diagonal blocks $-Q^{-1}A$ are not generally symmetric, so the full precision matrix is symmetric as a whole (as required) but its blocks are not individually symmetric. For the special case $A = \alpha I_d$ and $Q = \sigma_\eta^2 I_d$ (isotropic scalar-times-identity dynamics), everything reduces to d independent copies of the scalar AR(1) case and the structure is exactly as before, just replicated d times.

10 The Kalman Filter and Smoother

10.1 Why the Kalman smoother matters here

In the Mézard formula (Theorem 6.1):

$$S_k(\mathbf{x}, t) = \frac{e^{-t} \langle a_k \rangle_{\mathbf{a} | \mathbf{x}} - x_k}{\Delta_t},$$

the central object is $\langle a_k \rangle_{\mathbf{a} | \mathbf{x}}$: the posterior mean of the *clean* frame a_k given the entire *noisy* trajectory $\mathbf{x} = (x_0, \dots, x_K)$.

Computing this posterior mean is *exactly* the problem solved by the **Kalman smoother**. The Kalman smoother is therefore not merely a useful tool — it *is* the explicit propagator.

10.2 The Kalman filter: forward pass

The Kalman filter processes the noisy observations sequentially, maintaining a Gaussian belief over the current clean frame.

Initialisation. Prior belief at time 0:

$$p(a_0) = \mathcal{N}(a_0; \mu_0, \sigma_0^2).$$

Prediction step. Given the posterior at time k , $p(a_k | x_0, \dots, x_k) = \mathcal{N}(a_k; \hat{a}_{k|k}, P_{k|k})$, predict the next frame using the dynamics:

$$\hat{a}_{k+1|k} = \alpha \hat{a}_{k|k}, \quad P_{k+1|k} = \alpha^2 P_{k|k} + \sigma_\eta^2.$$

Update step. Upon observing x_{k+1} , compute the *Kalman gain*:

$$\mathcal{K}_{k+1} = \frac{e^{-t} P_{k+1|k}}{e^{-2t} P_{k+1|k} + \Delta_t},$$

and update the belief:

$$\hat{a}_{k+1|k+1} = \hat{a}_{k+1|k} + \mathcal{K}_{k+1} (x_{k+1} - e^{-t} \hat{a}_{k+1|k}), \quad P_{k+1|k+1} = (1 - e^{-t} \mathcal{K}_{k+1}) P_{k+1|k}.$$

The innovation $x_{k+1} - e^{-t} \hat{a}_{k+1|k}$ is the surprise in the new observation relative to the prediction. The Kalman gain $\mathcal{K}_{k+1}$ weights how much to update: large when the prediction uncertainty $P_{k+1|k}$ is large, small when the noise Δ_t is large.

10.3 The Kalman smoother: backward pass

The filter gives $\hat{a}_{k|k}$ — the best estimate using only observations up to time k . The *smoother* improves this by running a backward pass that incorporates future observations.

Smoother initialisation. At the last frame:

$$\hat{a}_{K|K}^s = \hat{a}_{K|K}, \quad P_{K|K}^s = P_{K|K}.$$

Backward recursion. For $k = K - 1, K - 2, \dots, 0$:

$$L_k = P_{k|k} \alpha P_{k+1|k}^{-1} \quad (\text{smoother gain}),$$

$$\hat{a}_{k|K}^s = \hat{a}_{k|k} + L_k (\hat{a}_{k+1|K}^s - \hat{a}_{k+1|k}), \quad P_{k|K}^s = P_{k|k} + L_k (P_{k+1|K}^s - P_{k+1|k}) L_k^\top.$$

The output $\hat{a}_{k|K}^s = \langle a_k \rangle_{a|x}$ is the exact posterior mean we need for the Mézard formula.

10.4 Connection to the joint score

The Kalman smoother provides a *recursive* way to evaluate $\mathbf{S}(\mathbf{x}, t)$ without ever explicitly forming or inverting the $(K + 1) \times (K + 1)$ matrix Σ_t . Instead, it computes $\langle a_k \rangle_{a|x}$ for all k in $\mathcal{O}(K)$ operations (two linear passes), versus $\mathcal{O}(K^3)$ for direct matrix inversion.

More importantly, the smoother *is* the propagator. The backward recursion step:

$$\hat{a}_{k|K}^s = \hat{a}_{k|k} + L_k (\hat{a}_{k+1|K}^s - \hat{a}_{k+1|k})$$

says: the smoothed estimate at frame k is updated from the smoothed estimate at frame $k + 1$ via the smoother gain L_k . This is the propagator P_t mapping $S_{k+1} \rightarrow S_k$ in closed form.

The Kalman smoother is the exact propagator. For the Gaussian AR(1) model, the hypothesised propagator $S_k \approx P_t(S_{k-1})$ is not an approximation — it is exact, and P_t is the Kalman smoother gain operator. Computing the joint score for the full $K + 1$ frame trajectory costs $\mathcal{O}(K)$ via the smoother, not $\mathcal{O}(K^3)$ via matrix inversion.

11 Suggested Visualisations

The following plots and animations are suggested to build intuition for the results in this document. Each targets a specific conceptual point.

11.1 Static plots

1. **Covariance vs precision matrix (side by side, heat map).** For $K = 4$, plot Σ_0 and Σ_0^{-1} as colour-coded grids with numerical values in each cell. Use distinct colours for diagonal, off-diagonal, and zero entries. Vary α with a slider to show how the dense Σ_0 compresses to diagonal while Σ_0^{-1} remains tridiagonal. *Key message: inversion extracts direct couplings.*

2. Σ_t^{-1} **entries vs diffusion time t** . Plot the diagonal entry $(\Sigma_t^{-1})_{kk}$ and the off-diagonal entry $|(\Sigma_t^{-1})_{k,k+1}|$ as functions of $t \in [0, 3]$. Mark the asymptotic values at $t = 0$ (tridiagonal limit) and $t \rightarrow \infty$ (diagonal limit). *Key message: inter-frame coupling in the score decays as e^{-2t} .*
3. **Joint score $S_0(x_0, x_1, t)$ vs x_0 for fixed x_1 , compared with marginal score.** For $K = 1$: plot both $S_0^{\text{joint}}(x_0, x_1, t)$ (blue) and $s^{\text{marg}}(x_0, t) = -(x_0 - \mu_{t,0})/T_{00}$ (red dashed) as functions of x_0 . Move x_1 with a slider. *Key message: the joint score shifts vertically with x_1 ; the marginal score is blind to it.*
4. **The $K = 2$ precision matrix at various t .** Four panels at $t = 0, 0.5, 1.0, 2.0$: show Σ_t^{-1} as a 3×3 grid. *Key message: off-diagonal entries shrink and matrix approaches I as t grows.*

11.2 Animations

1. **OU forward process on a trajectory.** Animate $K + 1$ frames of a clean AR(1) sample as t increases from 0 to 3: each frame independently drifts toward 0 and spreads, while the frames remain correlated (they all undergo the same shrinkage). Show the marginal distribution at each frame as a Gaussian envelope, and overlay sample paths. *Key message: noise is added independently per frame, but the trajectory remains correlated throughout.*
2. **Posterior mean (Kalman smoother) vs observation.** Fix a noisy trajectory $\mathbf{x}$. As t varies, animate how the Kalman-smoothed estimate $\langle \mathbf{a} \rangle_{\mathbf{a}|\mathbf{x}}$ moves. At $t = 0$: $\langle a_k \rangle = x_k e^t \approx x_k$ (perfect denoising). At large t : $\langle a_k \rangle \rightarrow \mu_{a,k}$ (prior dominates, observation is useless). *Key message: the score “denoising residual” shrinks as noise overwhelms the signal.*
3. **Joint score vector field on (x_0, x_1) plane, varying t .** For $K = 1$: quiver plot of (S_0, S_1) at each point of a grid. At $t \approx 0$: arrows are tilted off-axis (strong coupling). At large t : arrows point radially inward (score $\approx -\mathbf{x}/\Delta_t$, fully decoupled). Animate t continuously. *Key message: the propagator structure of the score fades continuously as t grows.*